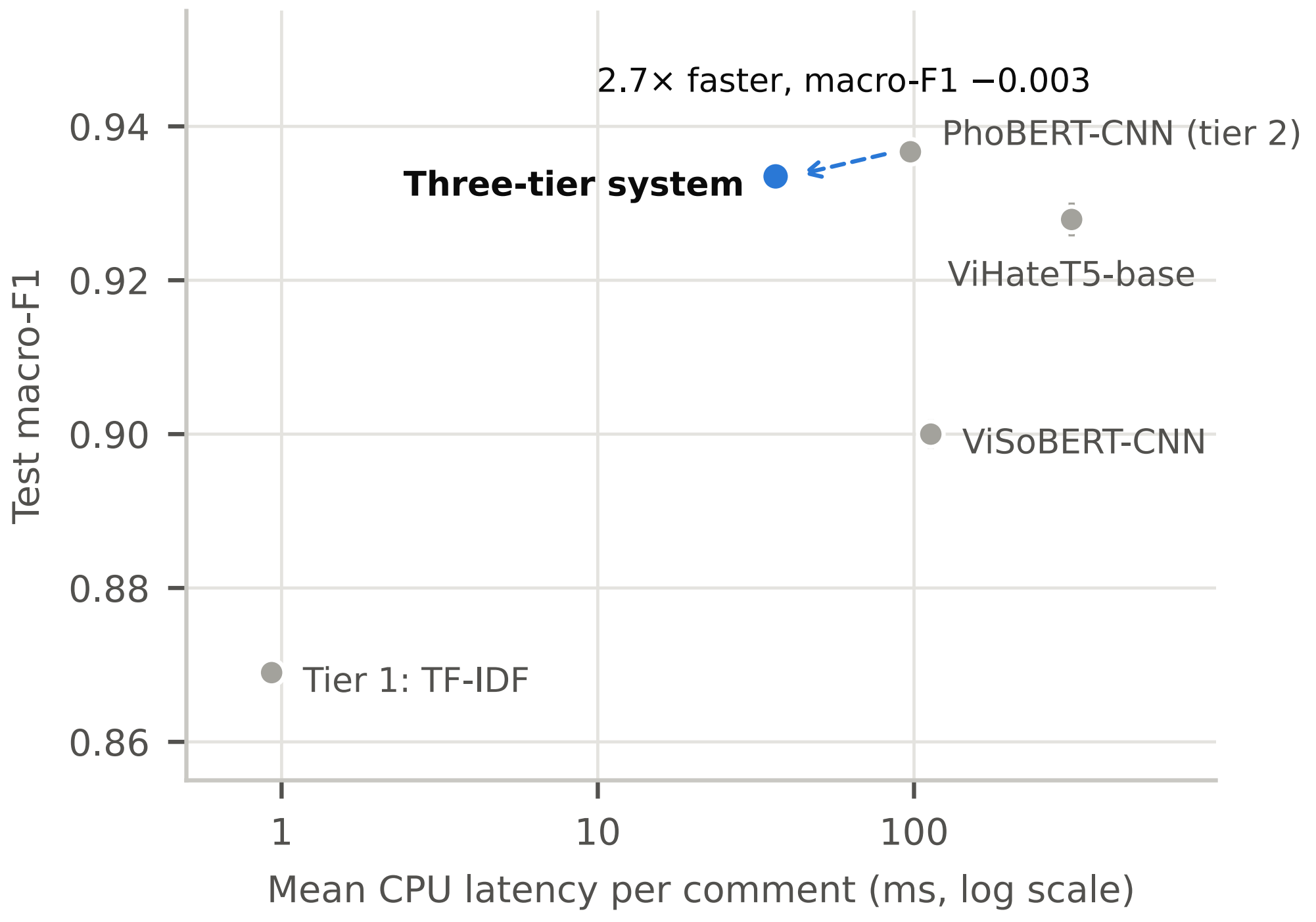
2.7× faster, macro-F1 −0.003
PhoBERT-CNN (tier 2)
Three-tier system
ViHateT5-base
ViSoBERT-CNN
Tier 1: TF-IDF
0.94
0.92
0.90
0.88
0.86
Test macro-F1
1
10
100
Mean CPU latency per comment (ms, log scale)